Next Prime

Time limit: 1.00 s
Memory limit: 512 MB

CSES 3396 | Mathematics | https://cses.fi/problemset/task/3396

Given a positive integer n, find the next prime number after it.

Input

The first line has an integer t: the number of tests.

After that, each line has a positive integer n.

Output

For each test, print the next prime after n.

Constraints

- $1 \le t \le 20$
- $1 \le n \le 10^{12}$

Example

Input
5 1 2 3 42 1337

Output
2 3 5 43 1361